

A

Industrial Oriented Mini Project Report on

# **REAL TIME OBJECT TRACKING WITH VISION BASED AI ANALYTICS**

*Submitted for partial fulfilment of the requirements for the award of the degree of*

**BACHELOR OF TECHNOLOGY**  
in the department of  
**COMPUTER SCIENCE AND ENGINEERING**  
by

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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**St. MARTIN'S ENGINEERING COLLEGE**

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**APRIL - 2026**



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## Certificate

This is to certify that the Industrial Oriented Mini project entitled “**Real Time Object Tracking With Vision Based AI Analytics**” is being submitted by S.Srujan Kumar (23K81A05P4), K.Sai Vignesh (23K81A05M7), M,Siddu (23K81A05N3),P.Minihas (23K81A05Q7) in the partial fulfilment of the requirement for the award of the degree of **BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING** is a bonafide work carried out by them. The result embodied in this report have been verified and found satisfactory.

Signature of the Guide  
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Signature of the HOD  
**Dr. K. SURESH**  
Professor and Head of the Department  
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**Internal Examiner**

**External Examiner**

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Place:

Date:



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## DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

### DECLARATION

We, the students of “**Bachelor of Technology** from the **Department of COMPUTER SCIENCE AND ENGINEERING**”, session: 2023 - 2027, **St. Martin's Engineering College, Dhulapally, Kompally, Secunderabad**, hereby declare that the work presented in this IOM project work entitled **Real Time object Tracking with Vision Based AI Analytics** is the outcome of our own bonafide work and is correct to the best of our knowledge and this work has been undertaken taking care of Engineering Ethics. This result embodied in this project report have not been submitted in any university for the award of any degree.

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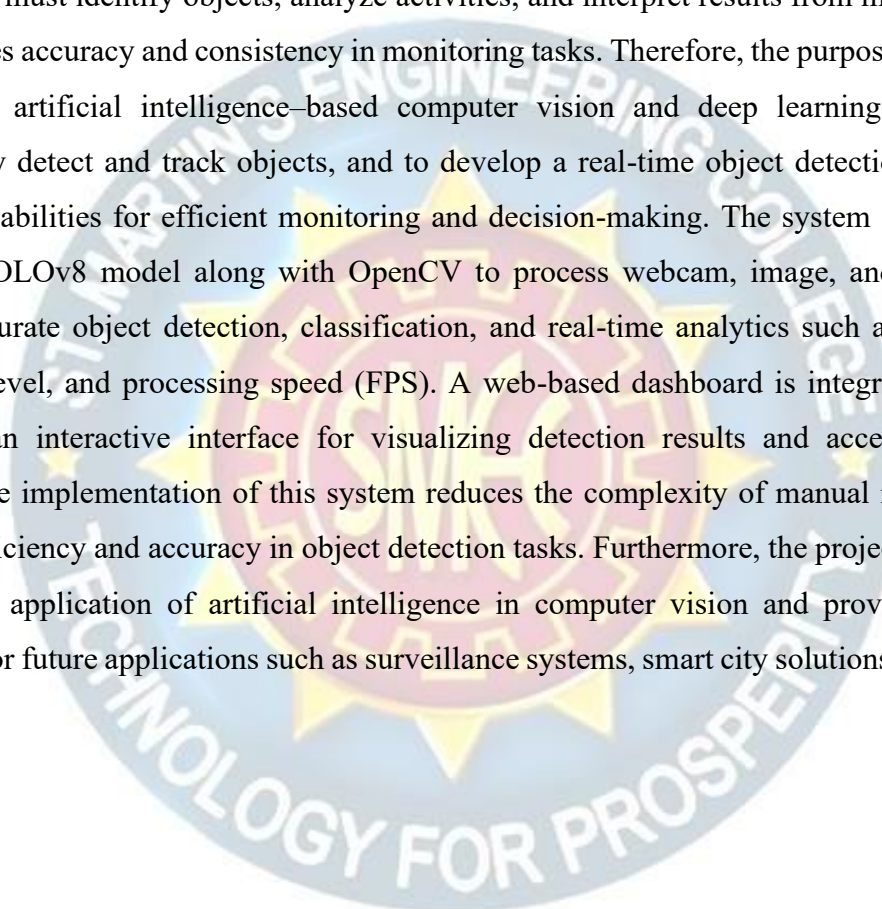
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## ABSTRACT

In many real-world environments, users rely on visual monitoring systems to observe and analyze objects from images, videos, or live camera feeds; however, traditional systems require continuous manual observation, making the process time-consuming, inefficient, and prone to human error. The operator must identify objects, analyze activities, and interpret results from multiple sources, which reduces accuracy and consistency in monitoring tasks. Therefore, the purpose of this project is to utilize artificial intelligence-based computer vision and deep learning techniques to automatically detect and track objects, and to develop a real-time object detection system with analytics capabilities for efficient monitoring and decision-making. The system is implemented using the YOLOv8 model along with OpenCV to process webcam, image, and video inputs, enabling accurate object detection, classification, and real-time analytics such as object count, confidence level, and processing speed (FPS). A web-based dashboard is integrated to provide users with an interactive interface for visualizing detection results and accessing analytics instantly. The implementation of this system reduces the complexity of manual monitoring and improves efficiency and accuracy in object detection tasks. Furthermore, the project demonstrates the practical application of artificial intelligence in computer vision and provides a scalable foundation for future applications such as surveillance systems, smart city solutions, and industrial automation.



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